Andrew Liu

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EDUCATION

University of California, Santa Barbara — Expected Graduation: June 2028

Bachelor of Science, Electrical Engineering — GPA: 3.98

RELEVANT COURSEWORK

Completed: Digital & Analog Circuits, Digital Design, Signals & Systems, Solid State Electronics, VLSI, Electromagnetics, Data Structures & Algorithms, Object-Oriented Programming (C++)

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern — June 2026 – September 2026

Hoplynk, San Francisco, CA

- Designed, debugged, and integrated power distribution & battery monitoring PCBs using COTS electronics under time constraints
- Created write-ups and documents to explain design choices, allowing for smooth design reviews
- Assembled and tested various electronics, including radios, using a multimeter and running test software

Avionics/Electrical Team Lead — January 2026 – Present

Gaucho Rocket Project, Santa Barbara, CA

- Design data acquisition systems and boards to log and stream rocket telemetry during flight & tests
- Coordinate and lead printed circuit board design workshops for new and current members
- Verify electronic components and sensor functionality with lab test equipment (multimeter, oscilloscope)
- Document test results and create protocols so the team has reference data and test instructions

Undergraduate Researcher — September 2026 – Present

Biomimetic Circuits & Nanosystems, UCSB, Advisor: Luke Theogarajan

- Studying an on-chip low-dropout regulator, which supplies a clean voltage to a voltage-controlled delay-locked loop
- Applied control theory to analyze the regulator's compensation as proportional-derivative feedback, re-deriving its block diagram

PROJECTS

Power Distribution & Battery Monitoring Printed Circuit Board — June 2026 – September 2026

Hoplynk, San Francisco, CA

- Designed a power distribution & battery charging board with DC inputs **(12V–32V)** + smart charger
- Used MOSFETs at IC enable pins to ensure pack-level isolation during battery discharge & charge
- Optimized board size while also derating all components to below 50% of max voltage for reliability
- Isolated high-power switching regions from sensitive analog measurements & digital lines through placement and grounding
- Implemented a digitally controlled load switch to allow cut-off of a heavy load, increasing system runtime by **∼86%**
- Wrote and flashed STM32 firmware and Python scripts to send and receive battery data & load-switch commands

Sensor Logger Printed Circuit Board V2 — October 2026

Gaucho Rocket Project, Santa Barbara, CA

- Designed a 4-layer PCB in KiCad to log temperature, pressure, & future add-on sensor data onto an SD card during rocket tests
- Broke out UART, SPI, & power rails to allow for future addition of other sensors & boards
- Stepped battery voltage to regulated rails using buck & boost converter ICs & added TVS protection on direct MCU pin paths
- Tested pressure transducer and protection circuits on perfboard, catching pinout issues before layout and avoiding a board respin

Boost Converter Printed Circuit Board — March 2026 – April 2026

Personal Project

- Designed and prototyped a boost converter circuit, achieving an efficiency of **∼66% (5V to 12V)**
- Implemented pulse-width modulation control with closed-loop voltage feedback using Arduino/ATtiny
- Designed PCB layout, taking into account trace width calculations and component placement

SUMMARY OF SKILLS

Technical: Analog and Digital Circuit Analysis, Circuit Prototyping and Debugging, KiCad, PCB Design & Rework, Soldering, Power Electronics, LTspice, C++, Python, Oscilloscope, Multimeter, Function Generator